



Performance Analysis of Data Link Layer Flow Control Protocols using NS-3

To implement, simulate, and comparatively analyze the performance of fundamental Data Link Layer ARQ protocols (Stop-and-Wait, Go-Back-N, and Selective Repeat) using the NS-3 network simulator.

Setup: Configure a simple point-to-point link topology in NS-3 with two nodes (sender and receiver). Utilize NS-3's point-to-point channel model and error model capabilities.

Implementation:

- Implement the logic for **Stop-and-Wait ARQ** between the two nodes.
- Implement the logic for **Go-Back-N ARQ**, including sender/receiver windows and cumulative ACKs.
- Implement the logic for **Selective Repeat ARQ**, including sender/receiver windows, buffering, and individual/selective ACKs/NAKs.

Simulation & Parameter Variation: Run simulations for each protocol under varying conditions:

- Vary the **channel Bit Error Rate (BER)**.
- Vary the **propagation delay** of the point-to-point link.
- Vary the **window size** (N for Go-Back-N, W for Selective Repeat).

Data Collection: Use NS-3's tracing and statistics framework to collect performance metrics, including:

- **Throughput (Goodput)** at the receiver
- **Average end-to-end packet delay**
- **Link utilization**
- **Number of retransmissions**

Expected Outcomes & Analysis:

Generate plots comparing **Throughput vs. BER**, **Delay vs. BER** and **Utilization vs. BER**. Generate similar plots showing the impact of **Propagation delay** and **Window size**.

Analyze the results, explaining the observed performance differences and trade-offs based on each protocol's mechanisms. Discuss conditions under which each protocol performs best or worst